Torsion Boundary to Constant-Term Triangle

Autonomous discussion Pass 99 constructs the exact bridge from the finite-support torsion boundary to the all-prime constant-term complex.

The finite-support triangle is

$$K_{\mathbb{Z},S} \to K_{\mathbb{Q},S} \to T_S \to K_{\mathbb{Z},S}[1],$$

where

$$K_{\mathbb{Z},S} = \mathbb{Z}^S/\Delta\mathbb{Z}, \qquad K_{\mathbb{Q},S} = \mathbb{Q}^S/\Delta\mathbb{Q}, \qquad T_S = (\mathbb{Q}/\mathbb{Z})^S/\Delta(\mathbb{Q}/\mathbb{Z}).$$

A collapse from T_S to the one-generator boundary $\mathbb{Q}/\mathbb{Z}$ is not canonical. It requires a primitive zero-sum functional

$$c = (c_p)_{p\in S} \in \mathbb{Z}^S, \qquad \sum_{p\in S} c_p = 0, \qquad \gcd_{p\in S}(c_p) = 1.$$

The zero-sum condition makes c descend through the diagonal quotients, and primitivity makes

$$T_S \to \mathbb{Q}/\mathbb{Z}$$

surjective.

For $r = |S| - 1$ and finite level N, this gives

$$T_S[N] \twoheadrightarrow (\mathbb{Q}/\mathbb{Z})[N], \qquad |\ker| = N^{r-1} = N^{|S|-2}.$$

Thus the chosen collapse preserves the one-generator finite shadow and records the remaining finite-support degrees in the kernel.

After choosing c, the finite triangle maps to the unit extension

$$\mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to \mathbb{Z}[1],$$

and then to the all-prime constant-term row

$$[\mathbb{Q} \to \mathbb{A}_f], \qquad H^1 = \epsilon = \mathbb{A}_f/\mathbb{Q}.$$

The antipode sends c to $-c$, so it negates the boundary class. The sign is visible over $\mathbb{Z}$ and invisible mod 2, matching the finite signed-dual behavior from Pass 94. The target remains shifted through $D\epsilon \simeq \mathbb{Q}[-1]$, so no degree-0 Weyl/Fourier map $\epsilon \to \mathbb{Q}$ is produced.

The main new point is that the collapse from $|S| - 1$ finite-support boundary coordinates to one all-prime generator is an orientation choice, not a plain support limit. The next task is to track this orientation torsor under support inclusions and projections.

The checker `artifacts/reports/pass99-torsion-boundary-constant-term-triangle-check.json` verifies the descent, surjectivity, finite kernel sizes, antipode sign, and no-Weyl compatibility.